Diagrama de clases (modelo de dominio y repositorios)

Clases principales con sus atributos y operaciones clave

DogCare

Pet
id, userId, name, breed
sizeCategory: small|medium|large|giant
ageYears, ageMonths, weightKg
humanAgeYears, notes

UserProfile
userId, displayName, email
photoUrl?, phone?
role: user|admin

PetRepository · ProductRepository
observePets(): Flow<List<Pet>>
addPet / updatePet / deletePet
observeProducts(): Flow<List<Product>>
addProduct / updateProduct / deleteProduct
validateCartItemsList(items)

CalculateDogAgeUseCase
calculate(years, months, size)
→ DogAgeCalculationResult
humanYearsEquivalent, stage
nutrition/health/exerciseAdvice

Conversation
id, participantUids[2]
participantNames
lastMessageText, lastSenderUid

Product
id, sellerUid, sellerName
name, brand, presentation
category: dry_food|wet_food|snacks
lifeStage, priceCop, stock
species = dog, imageUrl, active

OrderRepository · CartRepository
createOrder(...)
confirmApprovedPayment(order, ref)
markOrderRejected / markOrderError
observeUserOrders(): Flow<List<Order>>
cartSummary: StateFlow<CartSummary>
addToCart / updateQuantity / removeFromCart

EmergencyDetector
detect(text): Boolean
normaliza tildes y mayúsculas

ChatMessage
id, conversationId
senderUid, senderName
text, createdAt

1..*

PaymentGatewayRepository
processSandboxPayment(order, card, …)
→ PaymentReceipt (SIMULADO)
APPROVED | REJECTED | PENDING

CartItem / CartSummary
product: Product, quantity
subtotal
subtotalCop, shippingCop
totalCop, totalItemsCount

VetClinic
id, name, address, phone
latitude, longitude
is24hEmergency, openingHours
formatDistance(lat, lng)

AuthRepository · UserChatRepository
signUpWithEmail / signInWithEmail
signInWithGoogle / signOut
updateUserProfile(name, phone)
getOrCreateConversation / sendMessage
blockUser / reportUser

Order
id, userId, customerName
deliveryAddress, customerPhone
items: OrderItemRecord[]
quantities, subtotal/shipping/total
paymentStatus, paymentReference

AiChatMessage
id, isUser, text
isUrgent, isSystemError
canRetry, timestampMillis

AiAssistantRepository · VetClinicRepository
sendMessage(prompt, history)
MODEL_CANDIDATES (25 s por modelo)
búsqueda Overpass + respaldo Nominatim
openNavigationInGoogleMaps / dialPhoneNumber

Los repositorios (naranja) leen y escriben los modelos (verde) y los exponen como Flow/StateFlow.